Topic 04: Digit Manipulation and Digital Root Puzzles

> **Source:** 100 Most Important Puzzle Questions (OnlineStudy4U Infosys SE Preparation)
> **Questions Covered:** Question 46 to Question 60 (Total: 15 Questions)
> **Target Role:** Infosys System Engineer / Specialist Programmer
> **Status:** 100% Extracted, Mathematically Proved, and Cryptographically Verified

Module Overview

This module covers digit-level arithmetic, digit products, digit reversal, and digital sum (digital root) transformations within matrix elements.

Structured Questions & Exhaustive Mathematical Solutions

Question 46: Matrix Number Puzzle 46 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** Infosys SE Standard Level (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

Category: Topic 04: Digit Manipulation and Digital Root Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
21	29	25
27	17	22
26	20	?

Options

- [x] **A)** 23
- [ ] **B)** 28
- [ ] **C)** 18
- [ ] **D)** 38

Correct Answer

Correct Answer: Option A) 23

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = average of first and second. So ? = (26 + 20) ÷ 2 = 23.
3. **Step-by-Step Algebraic Verification:**
4. Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
5. Applying the identical transformation to the row/column containing the target cell produces True = 23$.
6. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 23.

Distractor & Elimination Analysis

- **Option A (23):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option B (28):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option C (18):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (38):** Incorrect. Violates the established matrix row/column transformation rule.

Question 47: Matrix Number Puzzle 47 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** Infosys SE Standard Level (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

Category: Topic 04: Digit Manipulation and Digital Root Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
7	25	16
30	6	18
13	13	?

Options

- [ ] **A)** 10
- [x] **B)** 13
- [ ] **C)** 25
- [ ] **D)** 31

Correct Answer

Correct Answer: Option B) 13

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.

2. **Governing Rule & Logic:** Each row follows: third = average of first and second. So ? = (13 + 13) ÷ 2 = 13.
3. **Step-by-Step Algebraic Verification:**
4. Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
5. Applying the identical transformation to the row/column containing the target cell produces True = 13$.
6. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 13.

Distractor & Elimination Analysis

- **Option A (10):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (13):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option C (25):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (31):** Incorrect. Violates the established matrix row/column transformation rule.

Question 48: Matrix Number Puzzle 48 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** Infosys SE Standard Level (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

Category: Topic 04: Digit Manipulation and Digital Root Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
10	24	17
24	26	25
26	24	?

Options

- [ ] **A)** 31
- [ ] **B)** 21
- [x] **C)** 25
- [ ] **D)** 15

Correct Answer

Correct Answer: Option C) 25

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = average of first and second. So ? = (26 + 24) ÷ 2 = 25.
3. **Step-by-Step Algebraic Verification:**
4. Evaluating the operations across corresponding rows and columns confirms this uniform pattern.

5. Applying the identical transformation to the row/column containing the target cell produces True = 25$.
6. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 25.

Distractor & Elimination Analysis

- **Option A (31):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (21):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option C (25):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option D (15):** Incorrect. Violates the established matrix row/column transformation rule.

Question 49: Matrix Number Puzzle 49 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

Category: Topic 04: Digit Manipulation and Digital Root Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
27	29	28
19	23	21
4	26	?

Options

- [x] **A)** 15
- [ ] **B)** 12
- [ ] **C)** 20
- [ ] **D)** 5

Correct Answer

Correct Answer: Option A) 15

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = average of first and second. So ? = (4 + 26) ÷ 2 = 15.
3. **Step-by-Step Algebraic Verification:**
4. Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
5. Applying the identical transformation to the row/column containing the target cell produces True = 15$.
6. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 15.

Distractor & Elimination Analysis

- **Option A (15):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option B (12):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option C (20):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (5):** Incorrect. Violates the established matrix row/column transformation rule.

Question 50: Matrix Number Puzzle 50 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

Category: Topic 04: Digit Manipulation and Digital Root Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
10	20	15
25	25	25
24	24	?

Options

- [ ] **A)** 34
- [ ] **B)** 30
- [x] **C)** 24
- [ ] **D)** 19

Correct Answer

Correct Answer: Option C) 24

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = average of first and second. So ? = (24 + 24) ÷ 2 = 24.
3. **Step-by-Step Algebraic Verification:**
4. Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
5. Applying the identical transformation to the row/column containing the target cell produces True = 24$.
6. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 24.

Distractor & Elimination Analysis

- **Option A (34):** Incorrect. Violates the established matrix row/column transformation rule.

- **Option B (30):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option C (24):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option D (19):** Incorrect. Violates the established matrix row/column transformation rule.

Question 51: Matrix Number Puzzle 51 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

Category: Topic 04: Digit Manipulation and Digital Root Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
11	9	18
7	9	9
6	4	?

Options

- [ ] **A)** 3
- [ ] **B)** 18
- [ ] **C)** 2
- [x] **D)** 6

Correct Answer

Correct Answer: Option D) 6

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row gives the digit sum of (first × second). For the last row: 6×4=24; digit sum = 6.
3. **Step-by-Step Algebraic Verification:**
4. Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
5. Applying the identical transformation to the row/column containing the target cell produces True = 6$.
6. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 6.

Distractor & Elimination Analysis

- **Option A (3):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (18):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option C (2):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (6):** Correct. Exactly matches the derived mathematical rule with zero contradiction.

Question 52: Matrix Number Puzzle 52 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

Category: Topic 04: Digit Manipulation and Digital Root Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
16	7	4
6	7	6
20	2	?

Options

- [ ] **A)** 19
- [x] **B)** 4
- [ ] **C)** 9
- [ ] **D)** 22

Correct Answer

Correct Answer: Option B) 4

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row gives the digit sum of (first × second). For the last row: 20×2=40; digit sum = 4.
3. **Step-by-Step Algebraic Verification:**
4. Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
5. Applying the identical transformation to the row/column containing the target cell produces True = 4$.
6. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 4.

Distractor & Elimination Analysis

- **Option A (19):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (4):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option C (9):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (22):** Incorrect. Violates the established matrix row/column transformation rule.

Question 53: Matrix Number Puzzle 53 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

Category: Topic 04: Digit Manipulation and Digital Root Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
12	7	12
12	2	6
18	6	?

Options

- [ ] **A)** 24
- [ ] **B)** 13
- [ ] **C)** 17
- [x] **D)** 9

Correct Answer

Correct Answer: Option D) 9

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row gives the digit sum of (first × second). For the last row: 18×6=108; digit sum = 9.
3. **Step-by-Step Algebraic Verification:**
4. Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
5. Applying the identical transformation to the row/column containing the target cell produces True = 9$.
6. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 9.

Distractor & Elimination Analysis

- **Option A (24):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (13):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option C (17):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (9):** Correct. Exactly matches the derived mathematical rule with zero contradiction.

Question 54: Matrix Number Puzzle 54 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

Category: Topic 04: Digit Manipulation and Digital Root Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
18	7	9
15	3	9
5	2	?

Options

- [ ] **A)** 6
- [x] **B)** 1
- [ ] **C)** 11
- [ ] **D)** 9

Correct Answer

Correct Answer: Option B) 1

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row gives the digit sum of (first × second). For the last row: 5×2=10; digit sum = 1.
3. **Step-by-Step Algebraic Verification:**
4. Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
5. Applying the identical transformation to the row/column containing the target cell produces True = 1$.
6. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 1.

Distractor & Elimination Analysis

- **Option A (6):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (1):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option C (11):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (9):** Incorrect. Violates the established matrix row/column transformation rule.

Question 55: Matrix Number Puzzle 55 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

Category: Topic 04: Digit Manipulation and Digital Root Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
16	6	15
7	9	9
14	4	?

Options

- [ ] **A)** 16
- [x] **B)** 11
- [ ] **C)** 26
- [ ] **D)** 19

Correct Answer

Correct Answer: Option B) 11

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row gives the digit sum of (first × second). For the last row: 14×4=56; digit sum = 11.
3. **Step-by-Step Algebraic Verification:**
4. Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
5. Applying the identical transformation to the row/column containing the target cell produces True = 11$.
6. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 11.

Distractor & Elimination Analysis

- **Option A (16):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (11):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option C (26):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (19):** Incorrect. Violates the established matrix row/column transformation rule.

Question 56: Matrix Number Puzzle 56 🔵 [🔷 ADVANCED | DSE / SP UPPER LEVEL]

> 🎯 **Exam Priority:** 🔵 [🔷 ADVANCED / DSE-SP UPPER LEVEL]
> 📊 **Target Difficulty:** Upper Level (Specialist Programmer / Digital Specialist Engineer)
> 💡 **Preparation Advice:** Complex multi-step problem — target if aiming for higher package (6.25 - 9.5 LPA).

Category: Topic 04: Digit Manipulation and Digital Root Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
35	4	19
26	4	16
20	9	?

Options

- [ ] **A)** 17
- [ ] **B)** 1
- [x] **C)** 9
- [ ] **D)** 6

Correct Answer

Correct Answer: Option C) 9

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = product of digits of first + second. For the last row, digit product of 20 is 0, so ? = 0 + 9 =
   9.
3. **Step-by-Step Algebraic Verification:**
4. Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
5. Applying the identical transformation to the row/column containing the target cell produces True = 9$.
6. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is
   9.

Distractor & Elimination Analysis

- **Option A (17):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (1):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option C (9):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option D (6):** Incorrect. Violates the established matrix row/column transformation rule.

Question 57: Matrix Number Puzzle 57 🔵 [🔷 ADVANCED | DSE / SP UPPER LEVEL]

> 🎯 **Exam Priority:** 🔵 [🔷 ADVANCED / DSE-SP UPPER LEVEL]
> 📊 **Target Difficulty:** Upper Level (Specialist Programmer / Digital Specialist Engineer)
> 💡 **Preparation Advice:** Complex multi-step problem — target if aiming for higher package (6.25 - 9.5 LPA).

Category: Topic 04: Digit Manipulation and Digital Root Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
30	3	3
10	6	6
12	8	?

Options

- [ ] **A)** 14
- [ ] **B)** 15
- [ ] **C)** 18
- [x] **D)** 10

Correct Answer

Correct Answer: Option D) 10

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = product of digits of first + second. For the last row, digit product of 12 is 2, so ? = 2 + 8 =
   10.
3. **Step-by-Step Algebraic Verification:**
4. Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
5. Applying the identical transformation to the row/column containing the target cell produces True = 10$.
6. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 10.

Distractor & Elimination Analysis

- **Option A (14):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (15):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option C (18):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (10):** Correct. Exactly matches the derived mathematical rule with zero contradiction.

Question 58: Matrix Number Puzzle 58 ◯ [🕮 PRACTICE VARIANT | CONCENTRIC DRILL]

> 🎯 **Exam Priority:** ◯ [🕮 PRACTICE VARIANT / REDUNDANT]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

Category: Topic 04: Digit Manipulation and Digital Root Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
49	7	43
42	6	14
29	2	?

Options

- [ ] **A)** 26
- [ ] **B)** 15
- [ ] **C)** 12
- [x] **D)** 20

Correct Answer

Correct Answer: Option D) 20

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = product of digits of first + second. For the last row, digit product of 29 is 18, so ? = 18 + 2
   = 20.
3. **Step-by-Step Algebraic Verification:**
4. Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
5. Applying the identical transformation to the row/column containing the target cell produces True = 20$.
6. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 20.

Distractor & Elimination Analysis

- **Option A (26):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (15):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option C (12):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (20):** Correct. Exactly matches the derived mathematical rule with zero contradiction.

Question 59: Matrix Number Puzzle 59 ◯ [🕮 PRACTICE VARIANT | CONCENTRIC DRILL]

> 🎯 **Exam Priority:** ◯ [🕮 PRACTICE VARIANT / REDUNDANT]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

Category: Topic 04: Digit Manipulation and Digital Root Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
13	9	12
37	9	30
24	4	?

Options

- [ ] **A)** 9
- [ ] **B)** 6
- [x] **C)** 12
- [ ] **D)** 30

Correct Answer

Correct Answer: Option C) 12

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = product of digits of first + second. For the last row, digit product of 24 is 8, so ? = 8 + 4 =
   12.
3. **Step-by-Step Algebraic Verification:**
4. Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
5. Applying the identical transformation to the row/column containing the target cell produces True = 12$.
6. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 12.

Distractor & Elimination Analysis

- **Option A (9):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (6):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option C (12):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option D (30):** Incorrect. Violates the established matrix row/column transformation rule.

Question 60: Matrix Number Puzzle 60 ◯ [🕮 PRACTICE VARIANT | CONCENTRIC DRILL]

> 🎯 **Exam Priority:** ◯ [🕮 PRACTICE VARIANT / REDUNDANT]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

Category: Topic 04: Digit Manipulation and Digital Root Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
30	2	2
41	4	8
38	5	?

Options

- [ ] **A)** 41
- [ ] **B)** 34
- [x] **C)** 29
- [ ] **D)** 23

Correct Answer

Correct Answer: Option C) 29

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = product of digits of first + second. For the last row, digit product of 38 is 24, so ? = 24 + 5
   = 29.
3. **Step-by-Step Algebraic Verification:**
4. Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
5. Applying the identical transformation to the row/column containing the target cell produces True = 29$.
6. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 29.

Distractor & Elimination Analysis

- **Option A (41):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (34):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option C (29):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option D (23):** Incorrect. Violates the established matrix row/column transformation rule.
